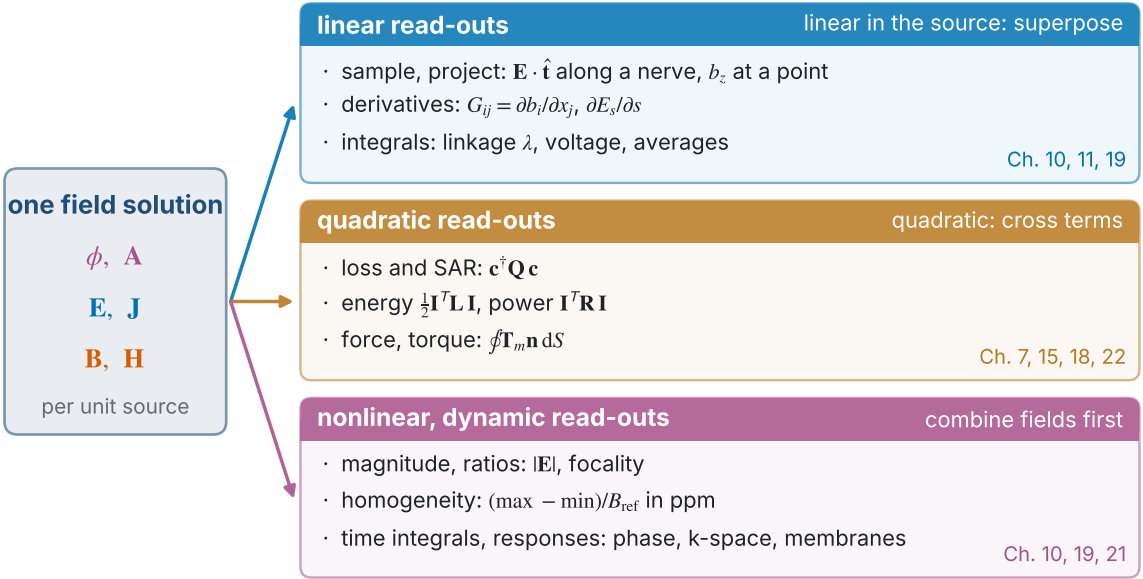
one field solution
ϕ, $\mathbf{A}$
$\mathbf{E}$, $\mathbf{J}$
$\mathbf{B}$, $\mathbf{H}$
per unit source
linear read-outs
linear in the source: superpose
· sample, project: $\mathbf{E}\cdot\hat{\mathbf{t}}$ along a nerve, b_z at a point
· derivatives: $G_{ij}=\partial b_i/\partial x_j$, $\partial E_s/\partial s$
· integrals: linkage λ, voltage, averages
Ch. 10, 11, 19
quadratic read-outs
quadratic: cross terms
· loss and SAR: $\mathbf{c}^{\dagger}\mathbf{Q}\,\mathbf{c}$
· energy $\frac{1}{2}\mathbf{I}^{T}\mathbf{L}\,\mathbf{I}$, power $\mathbf{I}^{T}\mathbf{R}\,\mathbf{I}$
· force, torque: $\oint\mathbf{T}_m\mathbf{n}\,\mathrm{d}S$
Ch. 7, 15, 18, 22
nonlinear, dynamic read-outs
combine fields first
· magnitude, ratios: $|\mathbf{E}|$, focality
· homogeneity: $(\max-\min)/B_{\mathrm{ref}}$ in ppm
· time integrals, responses: phase, k-space, membranes
Ch. 10, 19, 21